# ELHAM ROSTAMI

Doctoral Student, Université Paris-Saclay, Paris, France

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**Google Scholar**, **ResearchGate**, **LinkedIn**

## EDUCATION

**Doctoral Student (A MSCA fellow)** Oct 2025 - Now
*Université Paris-Saclay*, QS Ranking = 70, Paris, France, in Inria-saclay / CIAMS (BOOST team)

- PhD supervisor: Prof. Taous Meriem Laleg
- Co-supervisor: Prof. Hamidou Tembine
- Research topic: Improving the training, stability, and robustness of generative AI using control theory.
- Member of Marie Skłodowska-Curie Actions (MSCA)

**Master 2 in Computational Neurosciences and Neuroengineering (CNN)** Sep 2024 - Jul 2025
*Université Paris-Saclay*, QS Ranking = 70, Paris, France, following an internship in Inria-saclay (BOOST team)

- Academic advisor: prof. Sabir JACQUIR
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| Selected Courses | Grade |
|---|---|
| Machine learning | 19.8/20 |
| Methods and Stimulating | 17.5/20 |
| Dynamical System | 16.65/20 |
| Closed loop Neuroscience | 14.5/20 |

**MS.c. in Biomedical Engineering-Bioelectric (Avg.Grade=18.6/20, Ranked 2nd)** Oct 2018 - Sep 2021
*Amirkabir University of Technology (AUT)*, QS Ranking = 456, Tehran, Iran

- Ranked 2nd among MS.c. students
- Thesis: *Analysis of Electroencephalogram Signal for Offline Frequency Recognition in Brain Computer Interface Speller Based on Steady State Visually Evoked Potential.*
- Supervisor: Dr. Farnaz Ghassemi
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| Selected Courses | Grade |
|---|---|
| Machine Learning in Biomedicine | 18.5/20 |
| Pattern Recognition | 17.4/20 |
| Biological Signal Processing | 17/20 |
| Digital Signal Processing | 20/20 |
| Statistical Analysis of Medical Data | 20/20 |

**BS.c. in Biomedical Engineering-Bioelectric (Avg.Grade=18.19/20, Top %10)** Oct 2014 - Sep 2018
*Amirkabir University of Technology (AUT)*, QS Ranking = 456, Tehran, Iran

- Thesis: *Investigation of the Relationship between Mayers Briggs Personality Types and Internet Addiction.*
- Supervisor: Dr. Farnaz Ghassemi

**High School diploma (Avg. Grade=19.91/20)** Oct 2010 - Sep 2014
*Baharan School*, Ardabil, Iran

## RESEARCH INTERESTS

Generative AI, Optimal Control, Biomedical signal processing, Biomedical image processing, Brain–Computer Interface, Brain Signal Processing.

## PUBLICATIONS

### Accepted Papers

- **Rostami, E.**, Einizade, A., Laleg-Kirati, T.-M. (2026). *"SpaTeoGL: Spatiotemporal Graph Learning for Interpretable Seizure Onset Zone Analysis from Intracranial EEG"*, **EUSIPCO accepted**
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2023). *"Transfer Learning assisted PodNet for Stimulation Frequency Detection in Steady state visually evoked potential-based BCI Spellers"*, Brain-Computer Interfaces, 10, 38-49. (IF: 3.0)
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2022). *"Canonical Correlation Analysis of Task Related Components as a noise-resistant method in Brain-Computer Interface Speller Systems based on Steady-State Visual Evoked Potential"*, Biomedical Signal Processing and Control, 73, 103449. (IF: 5.1)
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2022). *"Improving the classification of real-world SSVEP data in brain-computer interface speller systems using deep convolutional neural networks"*, Frontiers in Biomedical Technologies, 9(4), 248-254. (IF: 0.5)
- **Rostami, E.**, Ghassemi, F., Tabanfar, Z. (2022). *"Improving the classification of Real-World SSVEP data in Brain-Computer Interface Speller systems using Deep Convolutional Neural Networks"*, In 5th Iranian Symposium on Brain Mapping Updates (ISBM), 8, 19-21.
- **Rostami, E.**, Ghassemi, F., (2020). *"Investigation of the effect of Attention in BCI systems based on SSVEP"*, In 8th Basic and Clinical Neuroscience Congress, Iran University of Medical Sciences.

### Papers Under review

- **Rostami, E.**, Laleg-Kirati, T.-M., Tembine, H. (2026). *"An Explicit Surrogate for Gaussian Mixture Flow Matching with Wasserstein Gap Bounds"*, arXiv preprint

## RESEARCH/JOB EXPERIENCE

- Internship in Inria-saclay, under the supervision of Prof. Taous-Meriem Laleg Kirati, **Project-team:** Bio-informed mOnitoring and Optimization for enhanced Sport and healTh (BOOST). Title: "Seizure Onset Zone Localization Using Visibility Graphs-Based Features." Jan 2025 - Jul 2025
- Senior Technical Mentor in AI: Python programming, Machine Learning, Computer Vision, AI Algorithms, *AIOLearn Academy*, Tehran, Iran. Feb 2023 - Jul 2023
- Judge and Deputy of Programming and Artificial Intelligence Competitions in the National Competitions of Creativity, Innovation, and Entrepreneurship, *AIOLearn Academy*, Tehran, Iran. Jun 2023 - Jul 2023
- Project Manager and Senior Programmer in "Personnel Identification System with Time Entry/Time Out", *AIOLearn Academy*, Tehran, Iran. May 2023 - Jun 2023
- Researcher at U: The Mind Company (development of a Message Passing Neural Network (MPNN) for Uncertainty Quantification), Columbus, Ohio. Jan 2022 - May 2022
- Internship: EEG signal recording and processing, National Brain Mapping Laboratory (NBML), Tehran, Iran. Jan 2022 - Feb 2022
- Research Assistant (RA): Brain Signal Processing, Revising Papers and Proposals, *Amirkabir University of Technology (AUT)*, Tehran, Iran. Dec 2021 - Jun 2022
- Teacher Assistant (TA): Research Methodology and Technical Writing, Computational Neuroscience Lab, *Amirkabir University of Technology (AUT)*, Tehran, Iran. Sep 2020 - Fab 2021
- Teacher Assistant (TA): Bioelectrical Phenomena, Action-potential Generation in excitable tissues and Electric Field Effects on the Body, Computational Neuroscience Lab, *Amirkabir University of Technology (AUT)*, Tehran, Iran. Sep 2019 - Jan 2020

## PROJECTS (For more information let me know, I will share the details with you)

### Academic Projects (Master and Bachelor)

- EEG signal quality analysis by Beacon Biosignals (The Dreem Headband), Score: 75.406%.
- Study on "Binocular Rivalry", Dynamical systems & Neuroscience, Universit´e Paris-Saclay.
- EEG signal Preprocessing, Feature extraction, and Classifications techniques
  *"specifically SSVEP signals, using the EEGLAB toolbox and EEG signal recording and processing"*
- Extraction of stimulation-induced Evoked Potentials from EEG signal by synchronous averaging
- Medical Engineering Internship at Mostafa Khomeini Hospital in Tehran (Operating Room Department)
  *"An internship in summer semester with forming valuable connections with doctors and nurses."*

## INVITED REVIEWER

**Conferences:** IEEE Conference on Decision and Control (CDC) and The European Signal Processing Conference (EUSIPCO).

## LICENSES AND CERTIFICATIONS

- National Competition of Innovation and Invention in Programming and AI, *AIOLearn Academy*, Tehran, Iran. Mar 2023
- Internship: EEG signal recording and processing, NBML, Iran. Jan 2022
- Applied Artificial Intelligence (based on Python), Tehran University, Iran. Jan 2019
- Brain Signal Processing in EEGLAB toolbox, Amirkabir University of Technology (AUT). Jan 2018

## COMPUTER SKILLS

**Programming Languages**
*Advanced:* Python, MATLAB.
*Intermediate:* HTML, CSS, PHP, SQL.

**Relevant Libraries**
Keras, Tensorflow, OpenCV, scikit-learn, EEGLAB toolbox, MATLAB GUI.

**Other Software and Tools**
Photoshop, LaTeX.

## LANGUAGES

**English:** Professional Sufficiency.
**Farsi:** Native Language.
**Turkish:** Native Language.
**French:** A1 level, currently in progress.

## AWARDS AND HONORS

- Distinguished student in master's degree, Biomedical Engineering Department, Amirkabir University of Technology, 2021.
- Ranked 2nd among MS.c. students in Biomedical Engineering Department, AUT, 2021
- Top %10 in BS.c. and Direct Admission to graduate school (M.Sc.) of Biomedical Engineering Department, AUT, without taking the national entrance exam, as a reward for academic records and achievements, 2018.
- Ranked Top %2 in the National University Entrance Exam among +222000 students, 2014.

## REFERENCES

**Associate Professor Taous-Meriem Laleg Kirati (Supervisor)**
*Electrical and Computer Engineering*
*INRIA Researcher, INRIA Paris-Saclay*
**Email:** *taous-meriem.laleg@inria.fr*

**Associate Professor Hamidou Tembine (Supervisor)**
*Department of Electrical Engineering and Computer Science*
*University of Quebec in Trois-Rivieres (UQTR), Quebec, Canada*
**Email:** *hamidou.tembine@uqtr.ca*

**Associate Professor Farnaz Ghassemi (previous Supervisor)**
*Department of Biomedical Engineering (Bioelectric)*
*Amirkabir University of Technology*
**Email:** *ghassemi@aut.ac.ir*

**Dr. Zahra Tabanfar Post Doc (Consultant, the percentage of cooperation $\geq 90\%$)**
*Technische Universität Dresden*
*Kognitive Neurophysiologie am Universitätsklinikum Carl Gustav Carus*
**Email:** *zahra.tabanfar@gmail.com*

**Mohammed (Mo) Abouelsoud**
*CEO at U: The Mind Company*
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